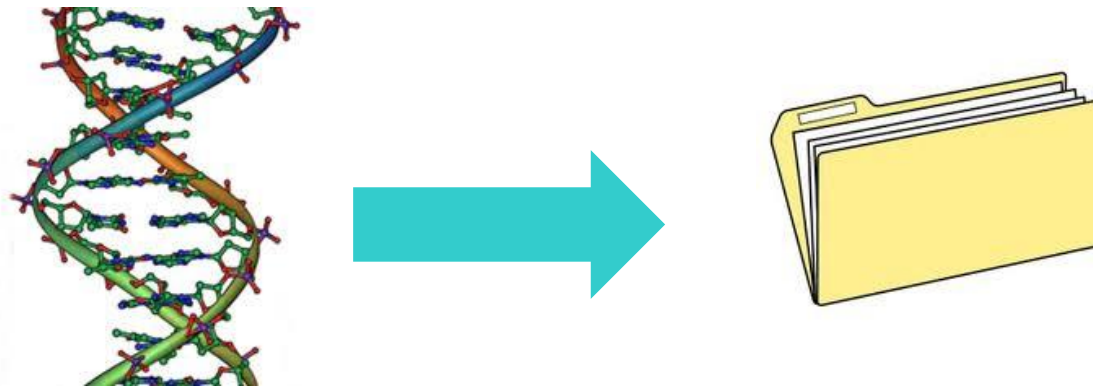


# DNA sequencing

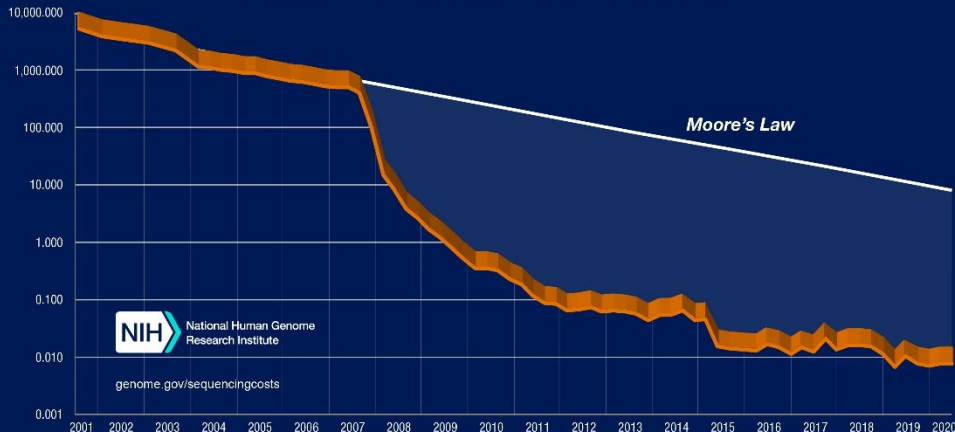
- Starting from a sample containing DNA, sequencing provides information about the exact sequence of the 4 bases in the molecule.



# Sequencing cost

- Sequencing cost is falling, while the value of the information is increasing
- At break-even, (1k\$?) sequencing could become a routine test for everyone.

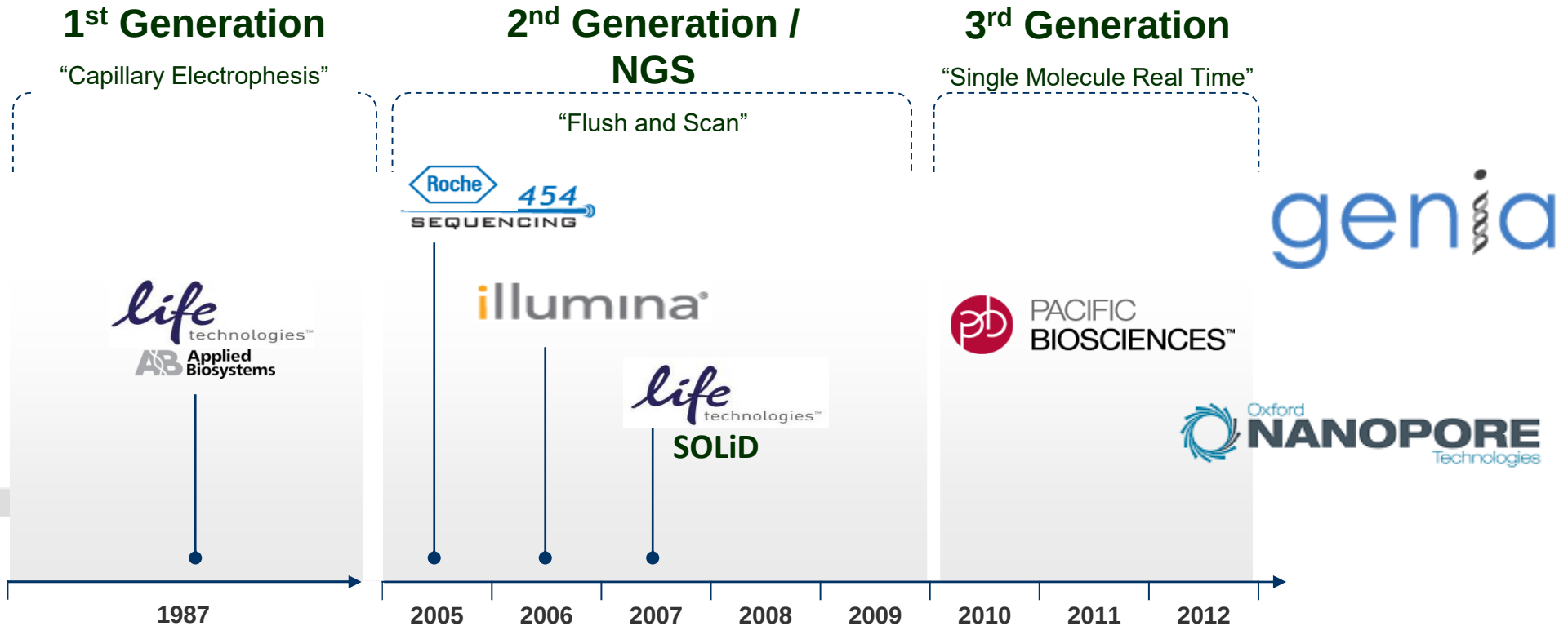
Cost per Raw Megabase of DNA Sequence



Cost per Human Genome

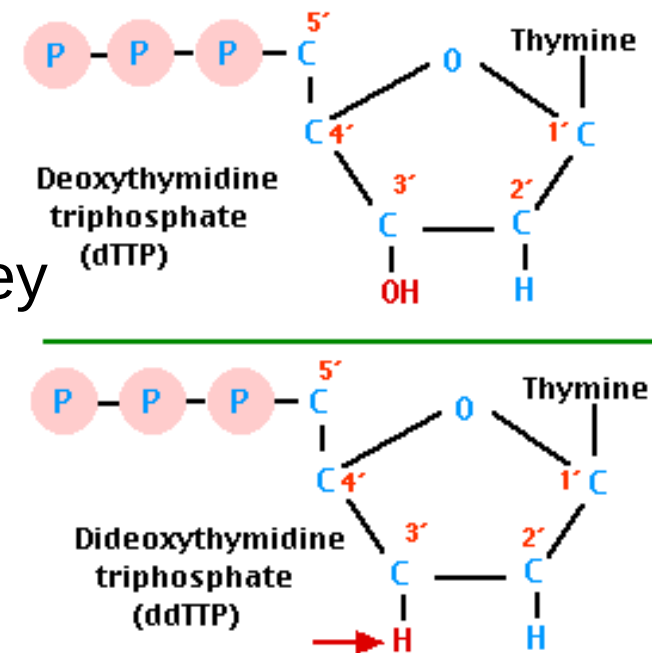


# Sequencing Platforms Launches & Companies



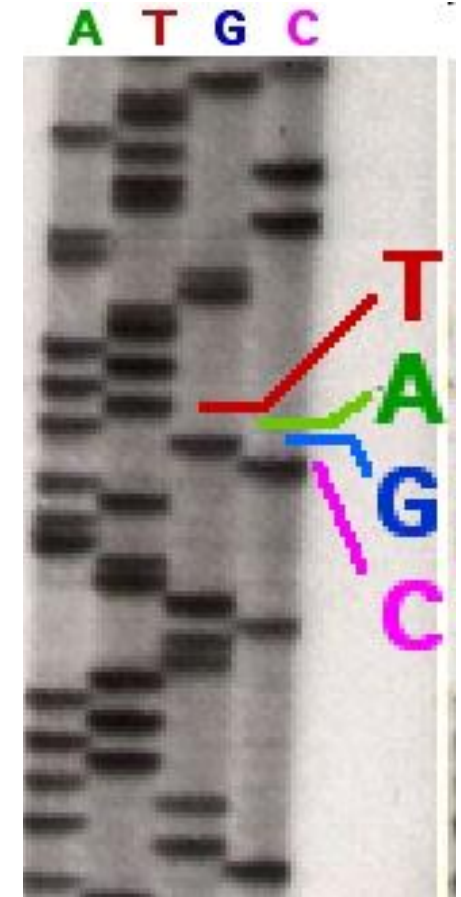
# First generation

- Sanger method (Chain termination “dideoxy”).
- In the PCR mix, beside the normal deoxy-tri-phosphate nucleotides, a small percentage of dideoxy-tri-phosphate nucleotides is introduced.
- Those modified bases are incorporated by the polymerase in the growing DNA, but they don't allow the next step.
- So, the amplified strands are interrupted at the point where the dideoxy has been incorporated



# Dna sequencing

- A small percentage of Dideoxy base of a single type (Ex. Thymine) is added to a PCR mix. The PCR is performed and the resulting DNA chains are interrupted where the complementary base was present ((Ex. Adenine)
- By electrophoresis it is possible to quantify the lengths of the fragments. The lengths corresponds to the places where Adenine is present
- Repeating 4 reactions, adding each time a different dideoxy (ddATP, ddTTP, ddCTP, ddGTP), it is possible to find the places of all the bases and so the sequence

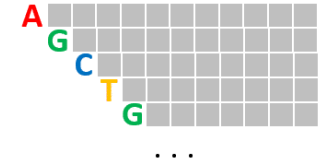


# Example:

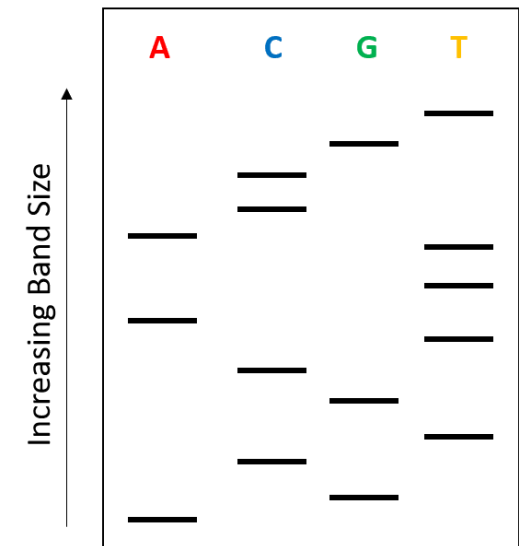
- DNA sequence: AGCTGCTATTACCGT...
- 1(ddATP)
  - A\*, AGCTGCTA\*, AGCTGCTATTA\*.
- 2(ddTTP)
  - AGCT\*, AGCTGCT\*, AGCTGCTAT\*, AGCTGCTATT\*, AGCTGCTATTACCGT\*.
- 3(ddGTP)
  - AG\*, AGCTG\*, AGCTGCTATTACCG\*.
- 4(ddCTP)
  - AGC\*, AGCTGC\*, AGCTGCTATTAC\*, AGCTGCTATTACC\*.

Sequence: A G C T G C T A T T A C C G T

Chain-terminating PCR:

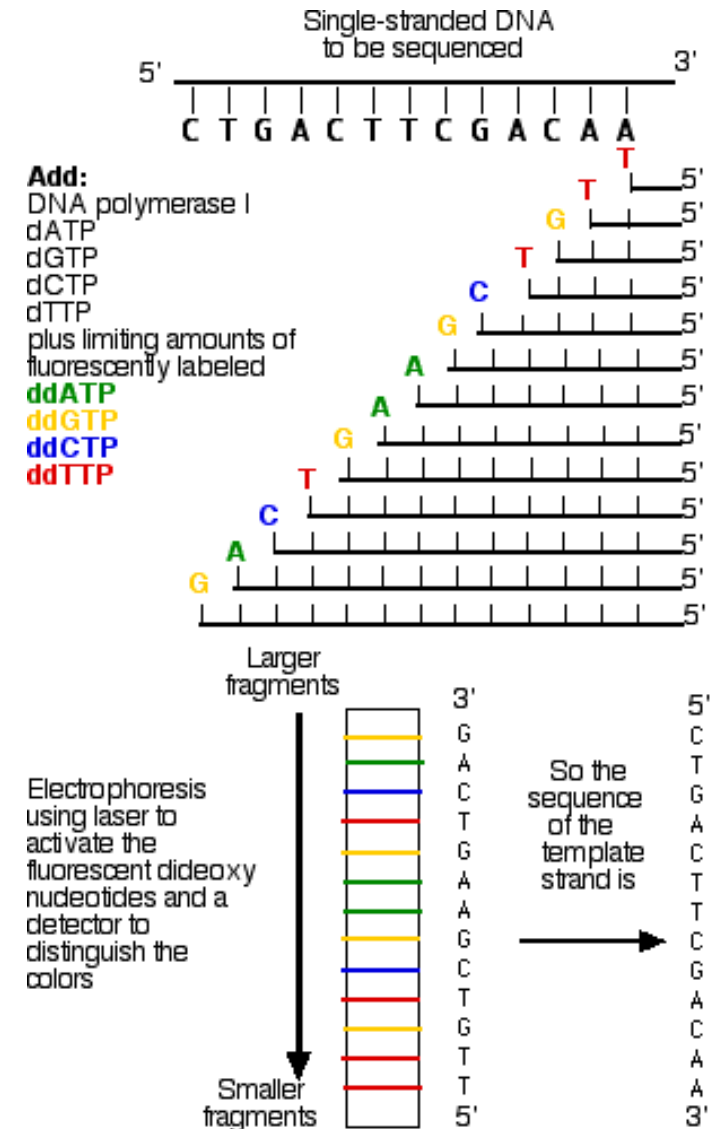


Gel Electrophoresis:



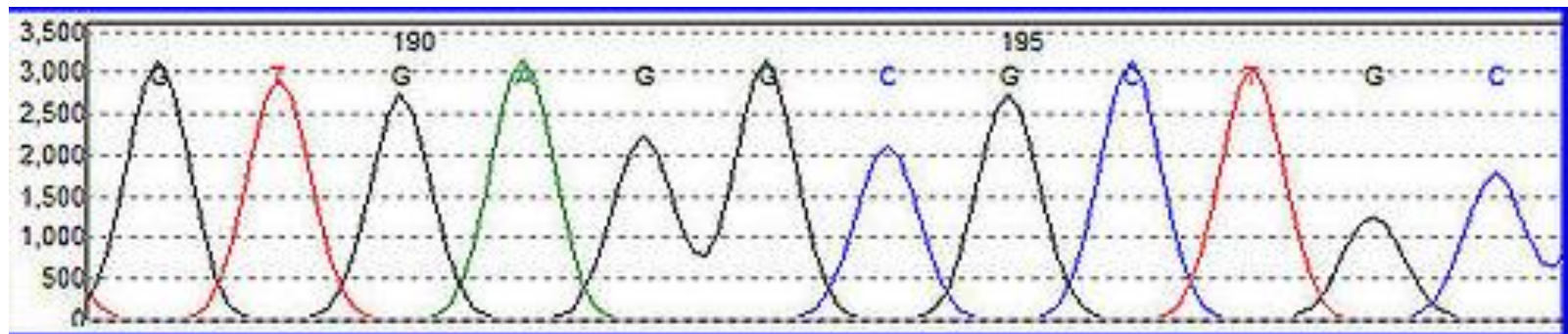
# Single reaction, 4 fluorescent channels

- The 4 dideoxy are labeled with different dyes at different wavelengths (different color)
- At every step a dideoxy may be incorporated, so all the possible lengths are present, but the fragment has the color corresponding to its last base.
- After amplification, a capillary electrophoresis with multicolor detection is performed.
- The color sequence at the detector is the DNA sequence



# Electropherogram

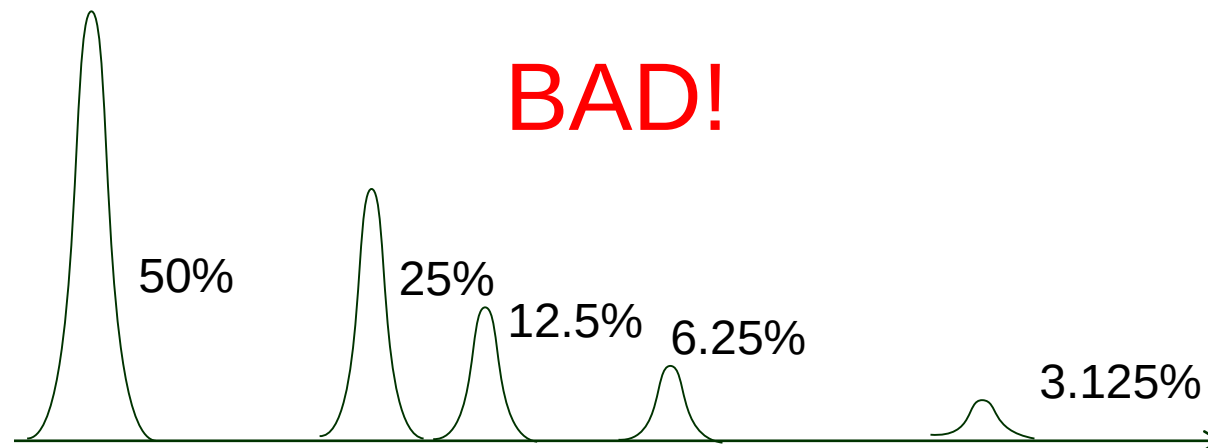
- Typical output of an automatic DNA sequencer with capillary electrophoresis and 4 colors detector.





# Percentage of dideoxy

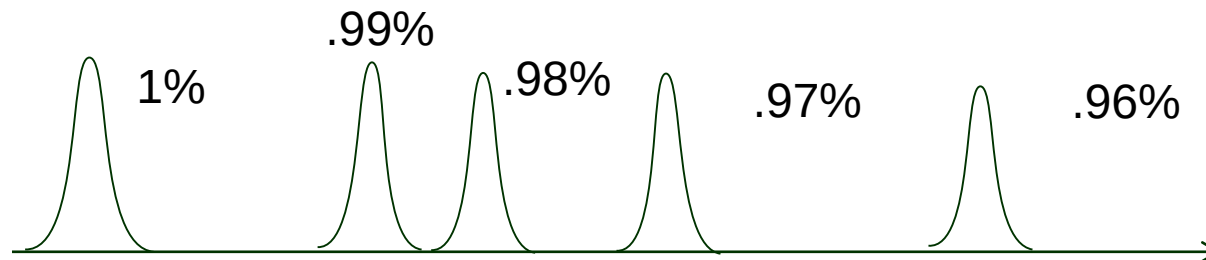
- The concentration of Dideoxy shall be much lower than that of normal Deoxy. Typically ( $<1\%$ ).
- If it is too high, too many chains are terminated immediately, leaving only few for the next step. This would limit the maximum length of the sequence.
- Es: 50%:



# Percentage of dideoxy

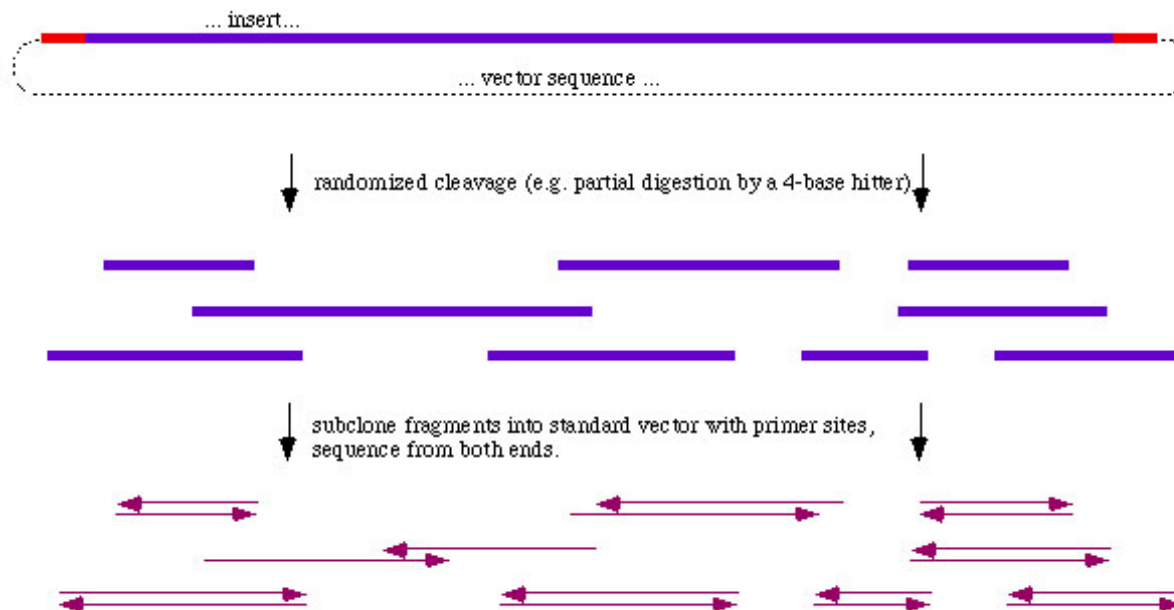
- The concentration of Dideoxy shall be much lower than that of normal Deoxy. Typically ( $<1\%$ ).
- If it is too high, too many chains are terminated immediately, leaving only few for the next step. This would limit the maximum length of the sequence.
- Es: 1%:

Good !



# Shotgun sequencing

- The whole genome is broken randomly in small segments.
- Starting from several copies of the genome, the fragments will partially overlap.
- Each segment is separately sequenced.
- After sequencing all the fragments, a software program finds the overlaps and matches the fragments reconstructing the whole sequence.



# Read-length Is Important

*Genome:*



*Second generation reads:*



*Genome:*



*Longer sequencing reads:*



# Second generation

- Flush and scan:

1. The 'library' is produced. It is a sample containing all the oligo subsequences generated by the shotgun specifically treated.

2. The library is used to make the array. Any oligonucleotide of the library is in a fixed place on the array, the position is random. The DNA is single strand and primed.

Preparation

Flush

3. A solution containing a pure base is flown on the array and a single base is made to be attached to the growing chain.

Run

Scan

4. It is measure where and if an incorporation happened
5. The procedure is repeated from Step 3, using a different base

6. For any pixel, the temporal series of measures gives the sequence of the corresponding oligonucleotide

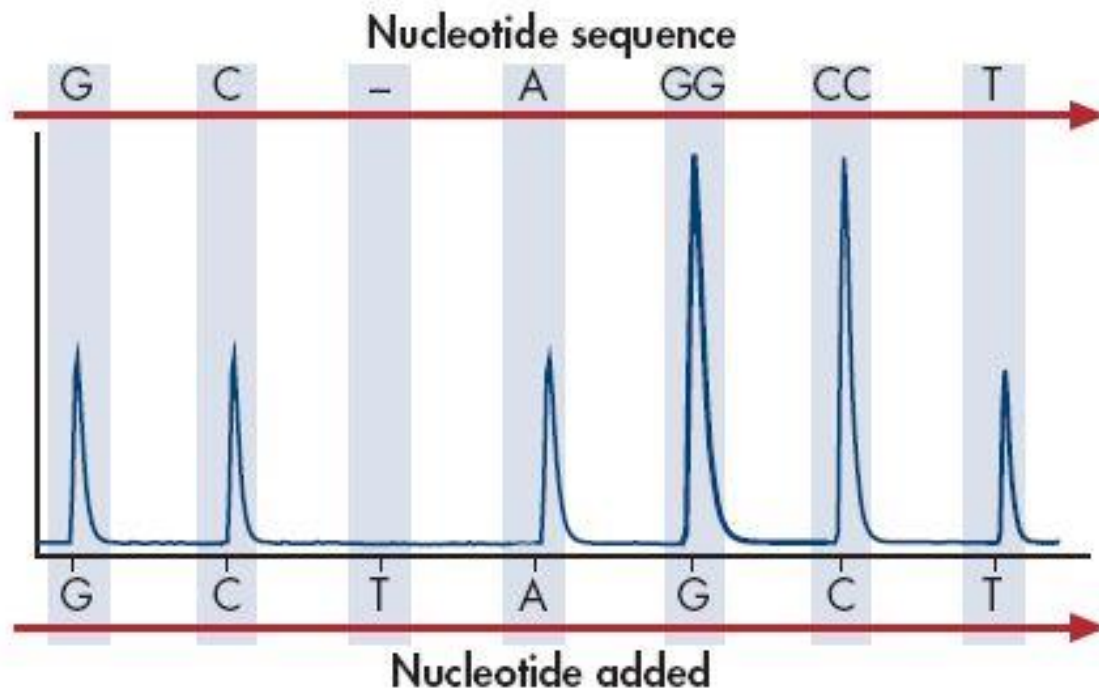
Post

7. Using Software the original whole sequence is reconstructed

Processing

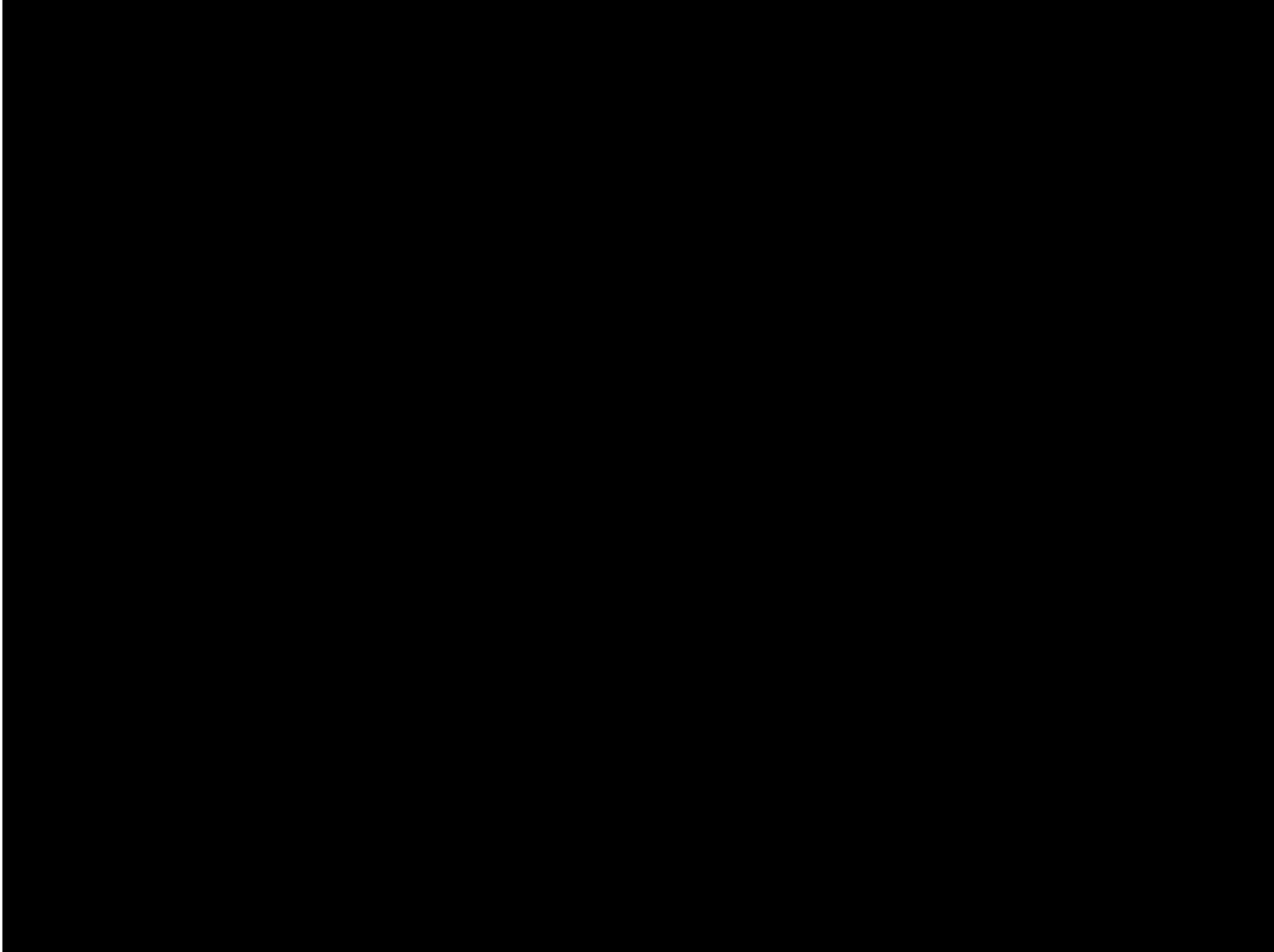
# Pyrogram

- For any single pixel is recorded a specific signal. This relates the sequence of insertion of bases to the sequence of DNA



# Solexa

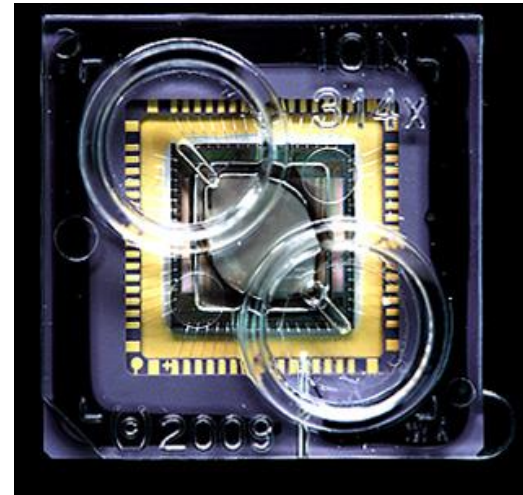
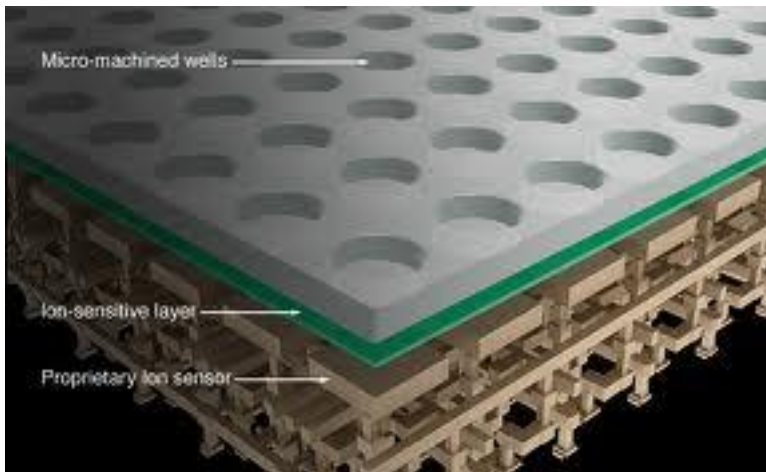
- Technology acquired by Illumina
- Currently (2016) is the most used system for DNA sequencing
- 4 Colors: each nucleotide marked with its own fluorophore
- Every flush step contains all the bases
- The places in the array are not at defined points
- <http://www.youtube.com/watch?v=I99aKKHcxC4&feature=endscreen&NR=1>





# Ion torrent Sequencing

- Newest technology acquired by Applied Biosystem (now Life technology, acquired by Thermo Fisher)
- Relies heavily on microelectronics technologies
- Electrical detection of the incorporation by measuring pH variation over the pixel
- <http://www.youtube.com/watch?v=yVf2295JqUg>



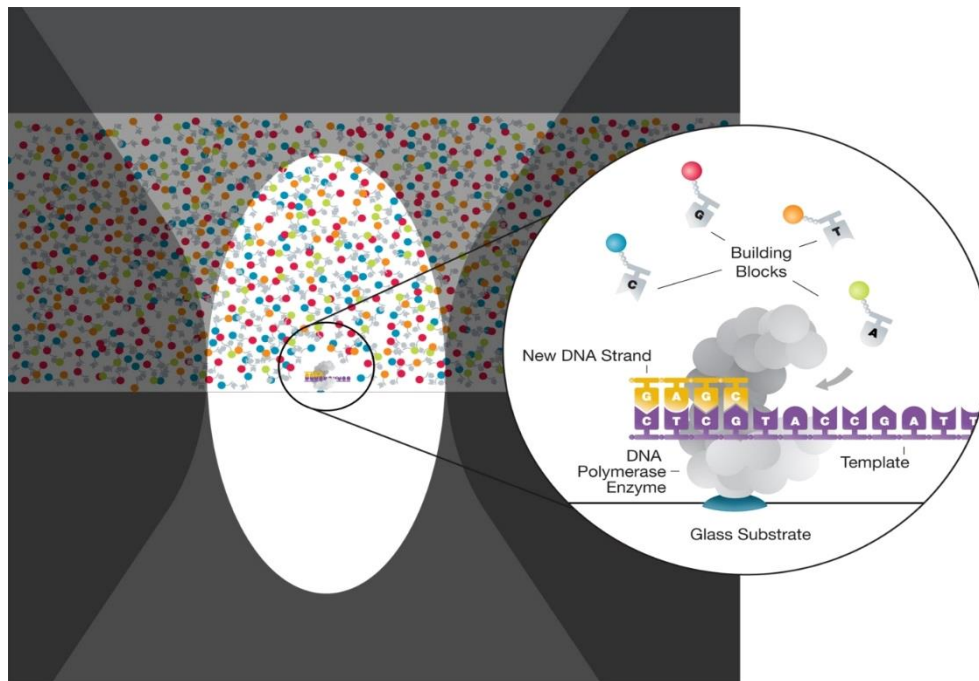


# Comparison of sequencing methods

|                        | Ion Torrent <a href="#">[14][16]</a> | 454 Sequencing <a href="#">[17]</a> | Illumina <a href="#">[18]</a>          | SOLiD <a href="#">[19]</a>                |
|------------------------|--------------------------------------|-------------------------------------|----------------------------------------|-------------------------------------------|
| Sequencing Chemistry   | Ion semiconductor sequencing         | <a href="#">Pyrosequencing</a>      | Polymerase-based sequence-by-synthesis | <a href="#">Ligation-based sequencing</a> |
| Amplification approach | Emulsion PCR                         | Emulsion PCR                        | Bridge amplification                   | Emulsion PCR                              |
| Mb per run             | 100                                  | 100                                 | 600,000                                | 170,000                                   |
| Time per run           | 1.5 hours                            | 7 hours                             | 9 days                                 | 9 days                                    |
| Read length            | 200 bp                               | 400 bp                              | 2x100 bp                               | 35x75 bp                                  |
| Cost per run           | \$ 350 USD                           | \$ 8,438 USD                        | \$ 20,000 USD                          | \$ 4,000 USD                              |
| Cost per Mb            | \$ 5.00 USD                          | \$ 84.39 USD                        | \$ 0.03 USD                            | \$ 0.04 USD                               |
| Cost per instrument    | \$ 50,000 USD                        | \$ 500,000 USD                      | \$ 600,000 USD                         | \$ 595,000 USD                            |

# PacBio (3rd generation)

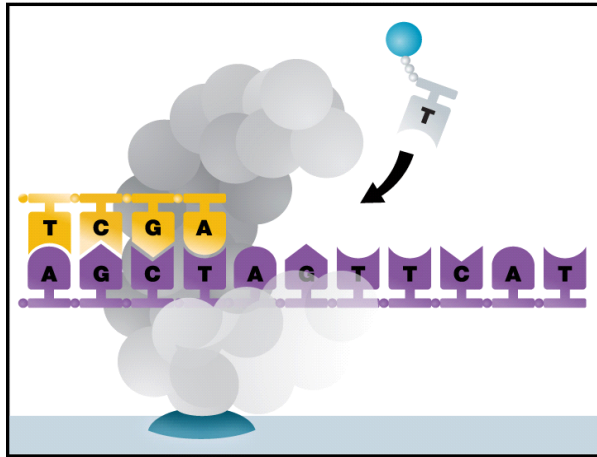
- DNA polymerase is placed on the bottom of a nano-well (1 single molecule per well)



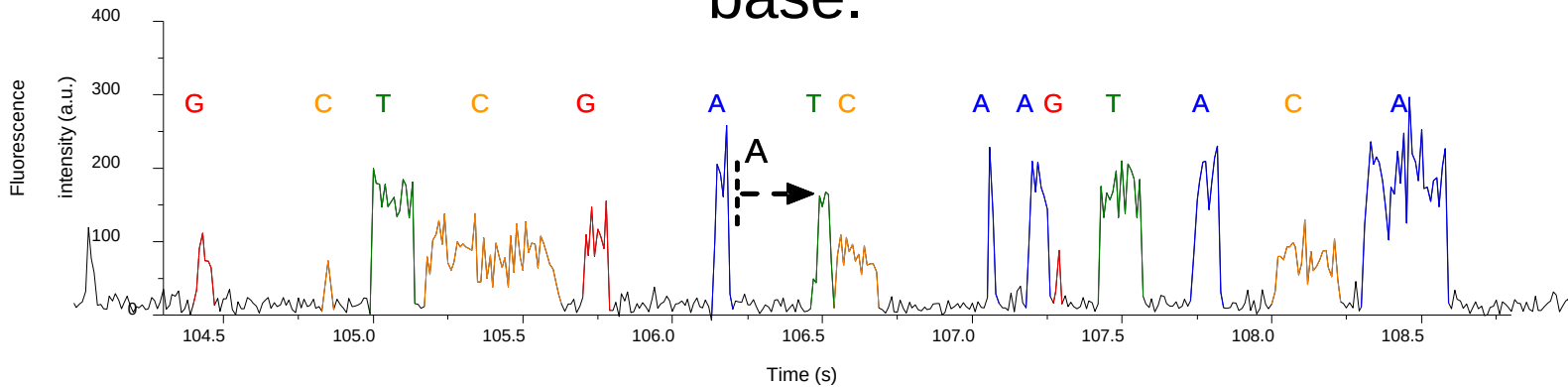
- The bases in solution are labelled with 4 different fluorophores
- Instead of being the polymerase moving on DNA is DNA moving over the fixed polymerase.



# Working principle



- Every time a new base is incorporated, the relative fluorophore emits a fluorescence flash, its color depends on the kind of added base.

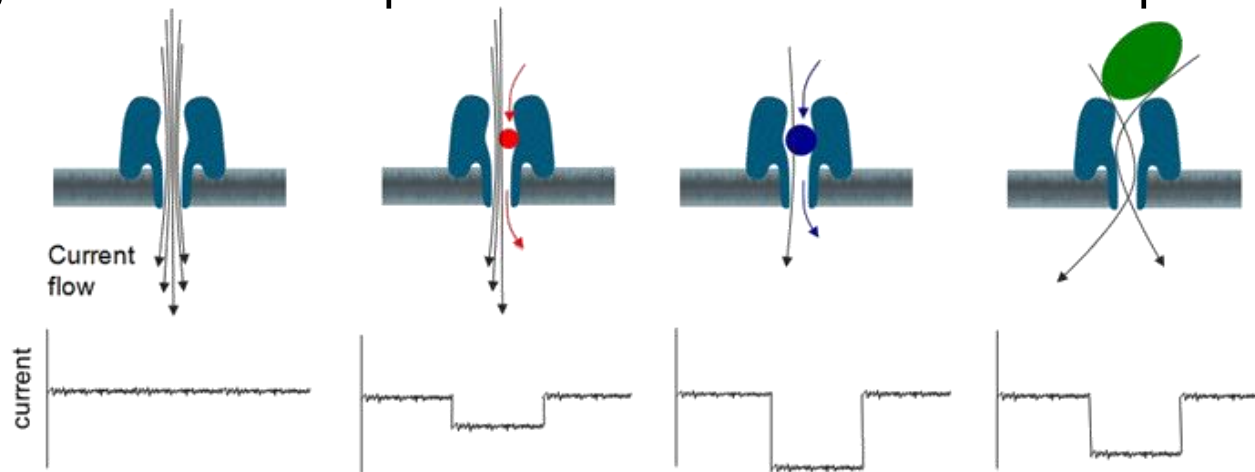


# Pac Bio



# Towards 4<sup>th</sup> generation: nanopores

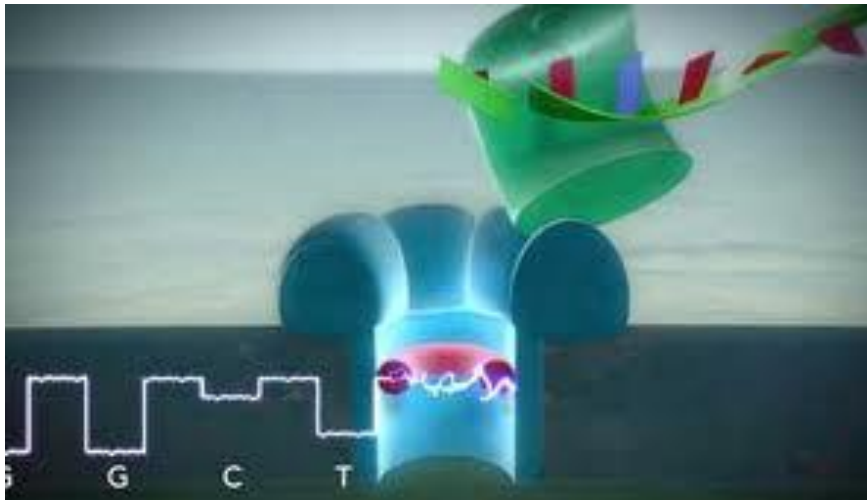
- A nanopore is a tiny hole in a membrane that separates two chambers containing ionic solution.
- A voltage is applied across the membrane, a current is produced by ions crossing the membrane through the hole
- When DNA enters in the pore, the ionic flow decreases and the measured current is reduced.
- Any macromolecule crossing the pore blocks the current in a different way, depending on its size.
- Measuring the current it is possible to know the size of the passing molecule

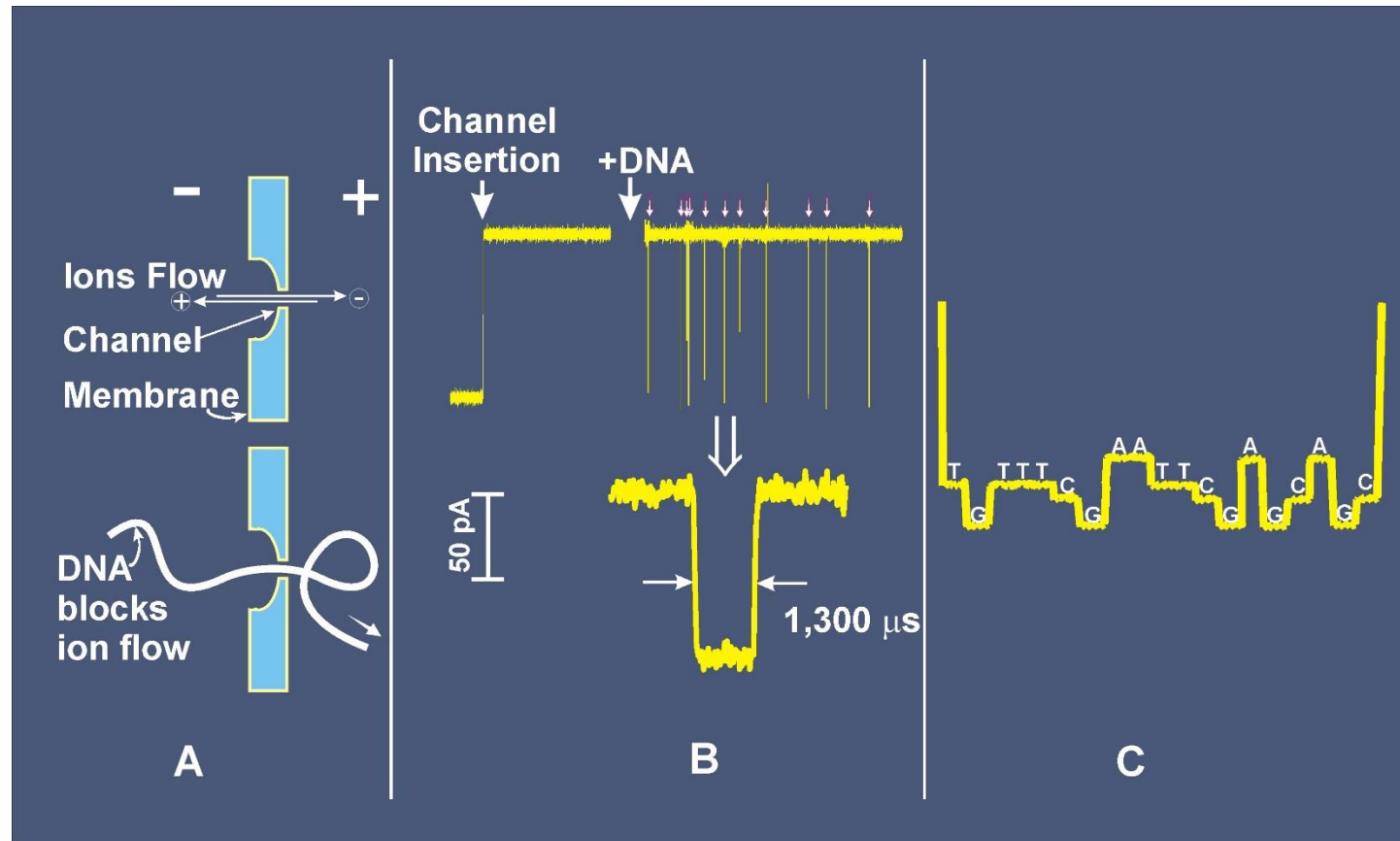


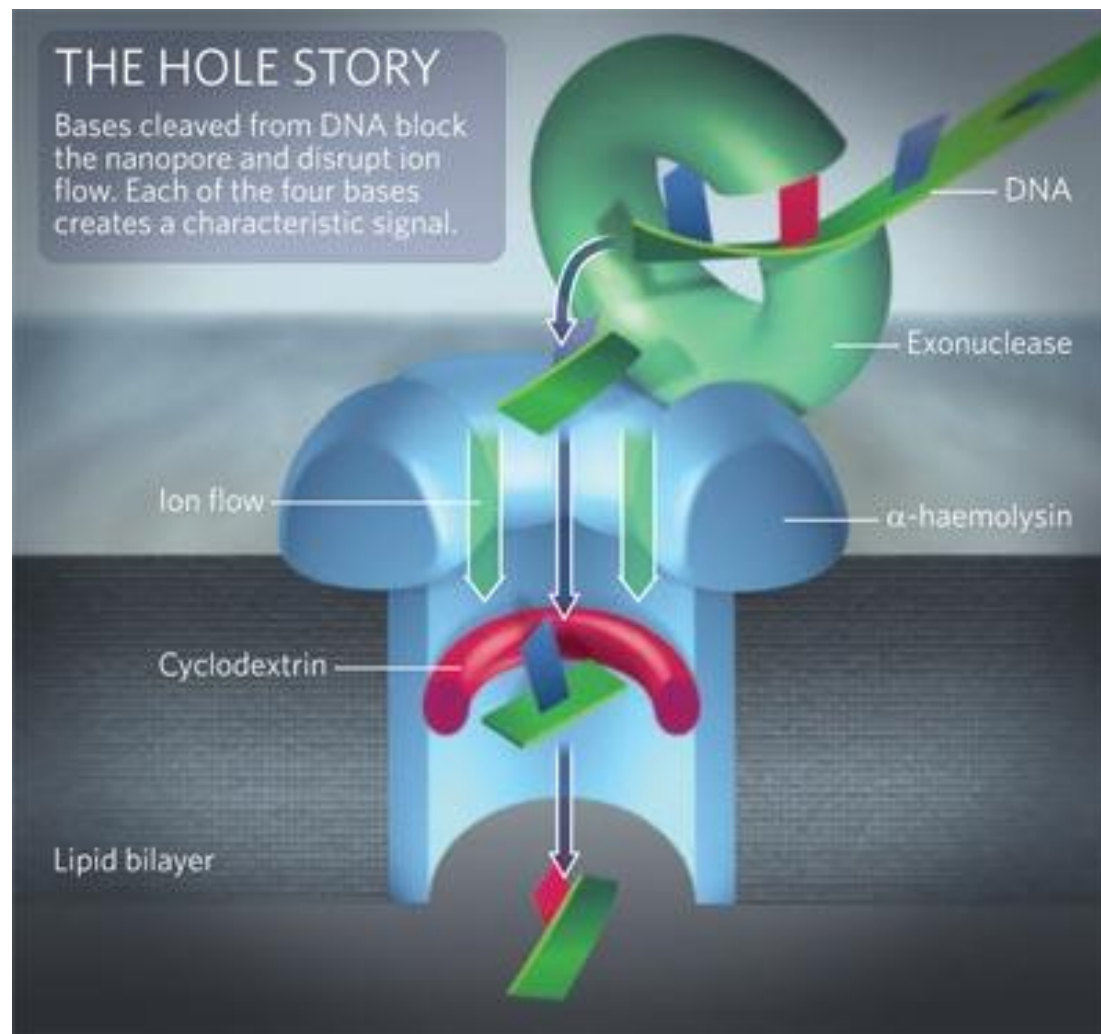


# Oxford nanopore

- Each nucleotide has a typical size, that produces a different current when passing through a nanopore. Measuring the current it is possible to know which base is passing by.



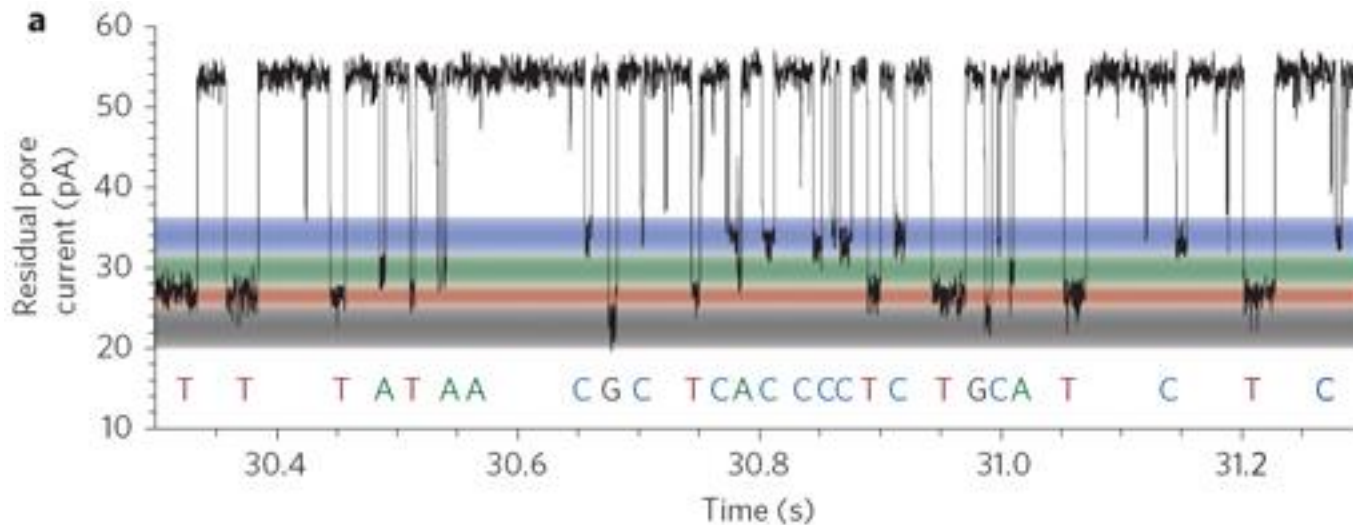




# MinION (Oxford nanopore)

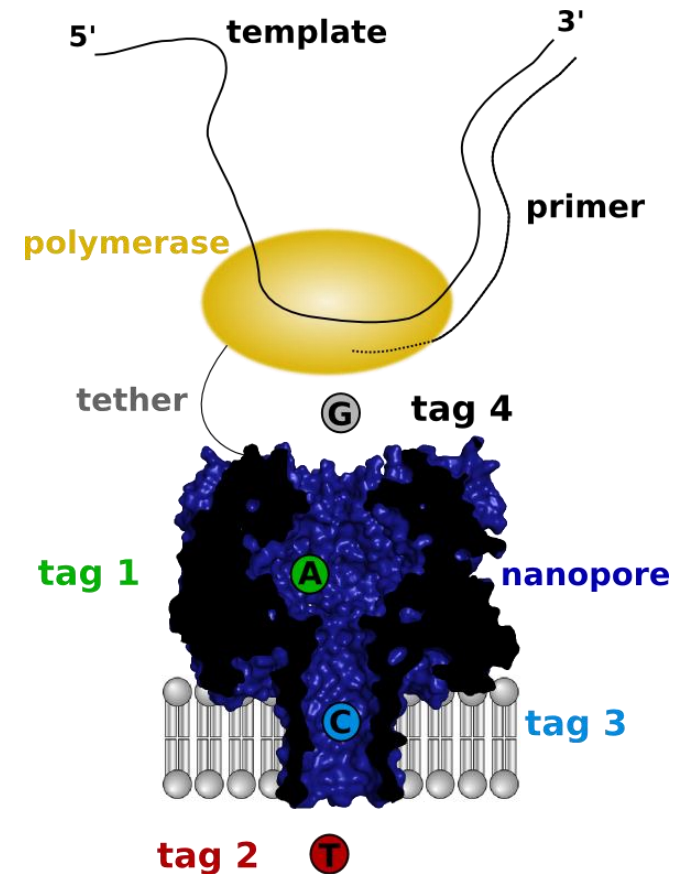


- Launched in 2015, just for beta testers in an evaluation program
- Enthusiastic reviews by users
- Still error rate > 20~30%



# Genia (Roche since 2014)

- A polymerase is tethered to the membrane close to the nanopore
- The triphosphate bases have a “tag” linked to the cleavable portion. Each type of base has a different tag.
- When a new base is incorporated, the tag+P+P is released.
- The tag passing through the pore, generates a signal.
- Compared to Oxford nanopore: since the tag is synthetic, it can be tailored to maximize the specificity of the signal.



# Sequencing by expansion

- Stratos genomics:  
<http://www.stratosgenomics.com/technology/>  
<http://www.youtube.com/watch?v=ADzIXItP4hE>
- A special polymerase translate the DNA into a different molecule, named 'expandimer'
- The expandimer contains the same information of the original DNA, but it is about 50 times bigger.
- Thanks to its dimensions, the expandimer is directly readable using standard techniques (for example, a bit on a hard-disk surface is smaller than few tens of nanometers)

